

A few methods to calculate limits:

First, try to use "direct substitution"

Example 1: Determine,

(a) $\lim_{x \rightarrow -1} (x^2 + x - 2)$

(b) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x + 2}$

$$\therefore \text{ (a) } \lim_{x \rightarrow -1} (x^2 + x - 2) = ((-1)^2 + (-1) - 2) = -2$$

$$\text{ (b) } \lim_{x \rightarrow 2} \frac{x^2 - 4}{x + 2} = \frac{(2)^2 - 4}{2 + 2} = \frac{0}{4} = 0$$

Note that, if $f(x) \equiv x^2 + x - 2$, $D_f = \mathbb{R}$ and

$-1 \in D_f$. Hence $\lim_{x \rightarrow -1} f(x) = f(-1) = -2$

similarly, if $g(x) = \frac{x^2 - 4}{x + 2}$, $D_g = \mathbb{R} \setminus \{-2\}$ and

$2 \in D_g$. Hence $\lim_{x \rightarrow 2} g(x) = g(2) = 0$

Example 2: $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$

$\therefore$ If using direct substitution we have

$$\frac{(-2)^2 - 4}{-2 + 2} = \frac{0}{0} \quad \text{Note that "0/0"}$$

is an indeterminate form. Direct substitution does not work !!!

Another method is "factorization".

$$\frac{x^2 - 4}{x + 2} = \frac{(x - 2)(x + 2)}{x + 2} = x - 2$$

Therefore, for example 2:

$$\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = \lim_{x \rightarrow -2} (x - 2) = -4$$

Observe that, if $f(x) = \frac{x^2 - 4}{x + 2}$, $D_f = \mathbb{R}^2 \setminus \{-2\}$ and $-2 \notin D_f$. Reason for the substitution not to be possible. But -2 is an element of some open interval $I \subset D_f$

One-sided limits

Definition 2: We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the left-hand limit of $f(x)$ as x approaches a [or the limit of $f(x)$ as x approaches a from the left] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and x less than a

Similarly, if we require that x be greater than a , we get "the right-hand limit of $f(x)$ as x approaches a is equal to L " and we write

$$\lim_{x \rightarrow a^+} f(x) = L$$

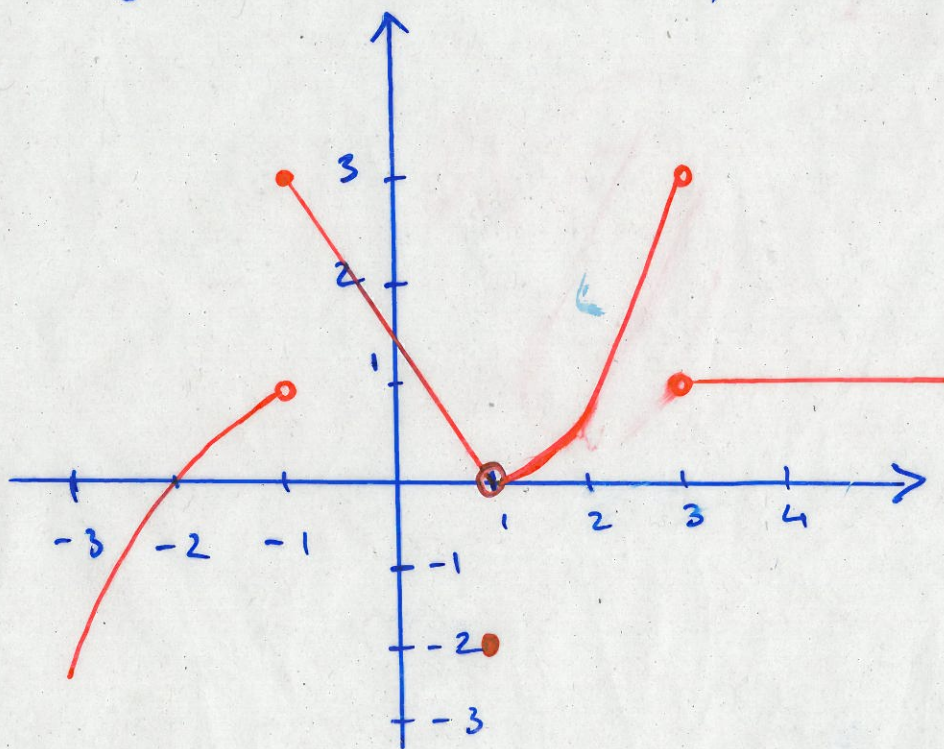
$\lim_{x \rightarrow a} f(x) = L$ if and only if

$$\lim_{x \rightarrow a^-} f(x) = L$$

AND

$$\lim_{x \rightarrow a^+} f(x) = L$$

Example 3: Use the graph of f to find the following limits (if they exist)



(a) $\lim_{x \rightarrow -1^-} f(x)$

(b) $\lim_{x \rightarrow -1^+} f(x)$

(c) $\lim_{x \rightarrow -1} f(x)$

(d) $\lim_{x \rightarrow 0^+} f(x)$

(e) $\lim_{x \rightarrow 0^-} f(x)$

(f) $\lim_{x \rightarrow 0} f(x)$

(g) $\lim_{x \rightarrow 1^-} f(x)$

(h) $\lim_{x \rightarrow 1^+} f(x)$

(i) $\lim_{x \rightarrow 1} f(x)$

(j) $\lim_{x \rightarrow 3^+} f(x)$

(k) $\lim_{x \rightarrow 3^-} f(x)$

(l) $\lim_{x \rightarrow 3} f(x)$

Example 3 :

- (a) $\lim_{x \rightarrow -1^-} f(x) = 1$ (c) $\lim_{x \rightarrow -1} f(x)$ does not exist
- (b) $\lim_{x \rightarrow -1^+} f(x) = 3$
- (d) $\lim_{x \rightarrow 0^+} f(x) = 1.5$ (f) $\lim_{x \rightarrow 0} f(x) = 1.5$
- (e) $\lim_{x \rightarrow 0^-} f(x) = 1.5$
- (g) $\lim_{x \rightarrow 1^-} f(x) = 0$ (i) $\lim_{x \rightarrow 1} f(x) = 0$
- (h) $\lim_{x \rightarrow 1^+} f(x) = 0$
- (j) $\lim_{x \rightarrow 3^+} f(x) = 1$ (l) $\lim_{x \rightarrow 3} f(x)$ does not exist.
- (k) $\lim_{x \rightarrow 3^-} f(x) = 3$

Example 4 : Find

$$\lim_{x \rightarrow 3^+} \sqrt{x-3}$$

and $\lim_{x \rightarrow 3^-} \sqrt{x-3}$

$$\lim_{x \rightarrow 3^+} \sqrt{x-3} = \sqrt{3^+ - 3} = \sqrt{0^+} = 0$$

(since $3^+ \in Df = [3, +\infty)$)

$\lim_{x \rightarrow 3^-} \sqrt{x-3}$ does not exist because there is no open interval $I \subset Df$ such that 3 is an element of I .

Example 5 : let $g(x) = \begin{cases} e^x, & x < -2 \\ x^2, & -2 \leq x \leq 2 \\ x+2, & x > 2 \end{cases}$

Find (a) $\lim_{x \rightarrow -2^-} g(x)$; (b) $\lim_{x \rightarrow -2^+} g(x)$

(c) $\lim_{x \rightarrow 2^-} g(x)$; (d) $\lim_{x \rightarrow 2^+} g(x)$

(e) $\lim_{x \rightarrow 2} g(x)$; (f) $\lim_{x \rightarrow 2} g(x)$

(h) $\lim_{x \rightarrow 1} g(x)$

Egs : solution :

$$(a) \lim_{x \rightarrow -2^-} g(x) = \lim_{x \rightarrow -2^-} e^x = e^{(-2)}$$

$$(b) \lim_{x \rightarrow -2^+} g(x) = \lim_{x \rightarrow -2^+} x^2 = (-2)^2 = 4$$

$$(c) \lim_{x \rightarrow 2^-} g(x) = \lim_{x \rightarrow 2^-} x^2 = (2)^2 = 4$$

$$(d) \lim_{x \rightarrow 2^+} g(x) = \lim_{x \rightarrow 2^+} (x+2) = 2+2 = 4$$

(e) from (a) and (b) $\lim_{x \rightarrow -2} g(x)$ does not exist

(f) from (c) and (d) $\lim_{x \rightarrow 2} g(x) = 4$

$$(h) \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} x^2 = (1)^2 = 1$$